



Field Survey Procedure - Fish Sampling

Restoration Monitoring Guidance

8 October 2025

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Acronyms and Abbreviations

CPUE	Catch-per-unit-effort
LWD	Large Woody Debris
PIT	Passive Integrated Transponder

1. OVERVIEW

Fish assessment methods vary depending on study objectives and the type of system in which sampling will occur. The procedures described in this guidance document are methods commonly used to survey fish populations at wadeable stream sites. It is important to consider which method(s) is/are most appropriate based on the study objectives and site conditions. When sampling paired restoration and control/reference sites, the same approach and, where possible, the same operators should be used to minimise variability.

Estimating the abundance of fish in a system is a common objective of sampling. Abundance estimates can be categorized as those focusing on either absolute or relative abundance. Absolute abundance refers to the total number of fish present in a specific area, whereas relative abundance provides an index of absolute abundance (e.g., CPUE; Quist et al. 2009). Techniques used to estimate absolute abundance include mark-recapture and depletion techniques. In larger habitats, these methods can be labour intensive and expensive, but may be appropriate depending on the species of interest and the project objectives. Alternatively, catch-per-unit-effort (CPUE; i.e., the number of fish sampled per unit of fishing effort) is often used as an index of abundance (Quist et al. 2009). In instances where catchability (i.e., the probability of catching an individual fish in one unit of effort) remains constant over time, trends in CPUE can accurately indicate changes in abundance (Hayes et al. 2007). However, estimates of absolute abundance are needed when catchability is likely to vary across space or time. In order to minimize uncontrolled sources of variation (error) in CPUE, study designs should incorporate standardization of gear and effort and identification of sampling times and locations (Hubert and Fabrizio 2007).

For monitoring the success of restoration projects, fish sampling will typically be required for several years before and over several years after the restoration actions, to identify patterns that are not due to random chance or associated with unknown sources of variability, and to account for the fish life-history timing. In addition, sampling at a control or reference site using the exact same sampling approach is strongly recommended. This allows you to control for causes of change in fish abundance that are not attributable to the restoration action (e.g., conditions at sea, at other locations in the watershed, parasitism/pathology, population dynamics, etc.).

Although the restored ('impact') site and the controls and/or references are chosen to be as similar as possible to each other, while remaining independent (i.e., fish do not move among them), they may still differ in their size. Similarly, a restoration project may result in a different magnitude of aquatic habitat available for fish, making the area of habitat from 'before' vs 'after' sampling different. As such, where it is possible to estimate or measure absolute abundance, it is often informative to convert the estimates of abundance (total number of fish in a habitat) to a standardised density metric (e.g., estimate of number of fish *per m²* of habitat). Although CPUE and density are both used to compare habitats / years, they are distinct because the former is per unit of effort (e.g., sets or hauls with a gear, the volume or area of habitat sampled, temporal duration of sampling) while the latter is per area – an important distinction for interpreting data.

Improved habitat may result in increases in fish density or abundance, and/or may result in improvements in biomass or condition. As an example, a channelised reach may have high CPUE values, moderate absolute abundance, high fish densities, but low average fork length. Restoring this to a stream-wetland complex may result in far lower CPUE values, a similar absolute abundance, and somewhat lower fish densities, but an indicator of success may be observed if fish have greater average fork lengths, consistently over time and in contrast to a control/reference site.

The sampling approach required will depend on the characteristics of fish populations that are of interest and the accuracy with which these must be estimated to meet the study objectives. Table 1 summarizes population characteristics that are commonly assessed and Table 2 provides an overview of methods used to assess fish population characteristics.

A summary of the required equipment for fish sampling is provided in Table 3.

Table 1: Frequently Assessed Fish Population Characteristics

Parameter or index	Type of Data	Description
Distribution	Presence/absence data	■ In habitats where fish are abundant or visibility is good, fish presence may be confirmed by visual observation. ■ If fish are not readily visible, apply methods that are effective at capturing the range of focal species and life stages in all of the types of habitat that are present. ■ More than one type of gear may be required if the habitat characteristics or fish life-stages vary widely.
Relative Abundance	Catch-per-unit-effort (CPUE)	■ An index of fish population abundance often used to assess spatial or temporal trends. ■ Relative abundance can be calculated from catch-per-unit-effort if catchability is equal between the entities that are being compared. ■ If catchability is likely to vary across time or space (due to differences in fishing gear, habitat, species etc.), then corrections may be necessary. This can be done by calibrating them with known estimates.
Absolute Abundance / Density	Area subsample, mark-recapture, or depletion	■ Absolute abundance is the number of fish present in a specific habitat, whereas density is the number of fish in a given area. ■ Knowledge of catchability is necessary to calculate absolute abundance. ■ Three techniques commonly used to estimate catchability include: ○ Mark-recapture methods – A known number of individuals are marked and released into the sampling area. The population is then sampled and its size estimated based on the ratio of marked and un-marked individuals. ○ Depletion Methods – Observe the rate at which catches decline with repeated sampling. ○ Known catchability, or calibrated methods – Use a pre-determined formula based on knowledge of gear efficiency/catchability ■ There are assumptions for each of the methods of estimating absolute abundance. These assumptions are critical to the accuracy of the estimates.

Sources: Portt et al. 2006, Pope et al. 2010

In addition to the types of data listed in Table 1, it may be of interest to examine characteristics of the fish community or focal population. For example, where a structure presents a partial barrier to fish passage, it may be of interest to sample upstream and downstream of the structure, comparing attributes like species composition, life-stages, or mean population fork length.

Table 2: Overview of Common Sampling Methods

Method	Assumptions	Application	Appropriate Habitat Conditions	Field Considerations	Limitations/Benefits
Catch-per-unit-effort sampling (CPUE; e.g., single-pass electrofishing)	■ Capture efficiency is independent of field conditions (e.g., discharge, temperature, depth, habitat complexity, turbidity). ■ The number of fish captured is proportional to the amount of effort expended	■ Determine fish presence or absence. ■ Determine fish species composition. ■ Can be used to determine an index of abundance. ■ Can be used to sample multiple sites within a limited timeframe.	■ Homogenous habitat (less complexity)		■ Cost effective ■ Least labour-intensive method for indexing abundance ■ Less precise than multiple pass/mark-recapture methods. ■ Poor estimator with complex cover (increased habitat complexity) ■ Poor estimator with increasing stream size ■ Low power detecting small changes in abundance ■ Highly variable data
Depletion Sampling (e.g., Multiple-pass removal)	■ Emigration and immigration by fish during the sampling period must be negligible ■ Vulnerability to capture of fish in a specified sample group must remain constant for each pass (e.g., fish do not become more wary of capture) ■ Equal fishing effort on all passes ■ All fish within a specified sample group must be equally vulnerable to capture during a pass.	■ Can be used to sample multiple sites within a relatively short timeframe. ■ Can be used to determine fish abundance or density	■ Low velocity streams (limited by block nets)	■ Complete a minimum of three passes or, for depletion sampling, until catch significantly drops and levels off. ■ Use block nets to prevent fish movement. ■ Equal effort applied for each pass ■ Decreasing catch on each pass	■ More precise than single pass ■ Capture probabilities among passes may vary due to behavioural avoidance ■ Variation in stream size or instream cover bias comparisons between sites

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	■ Conditions which affect collection efficiency (e.g., water clarity), must remain constant			■ 1 hour rest between passes to reduce gear avoidance ■ Collect site-specific habitat data (e.g., width, depth, gradient, thalweg depth, cover type, water velocity, temperature, clarity)	
Mark-recapture	■ Marked and unmarked fish are equally vulnerable to capture ■ Marked and unmarked fish have the same mortality rates ■ Marked fish randomly mix with unmarked fish ■ Marks are retained during the sampling period and all marks on recaptured fish are recognized ■ There is negligible emigration or immigration during the recapture period	■ Can be used to determine fish abundance or density	■ Low velocity streams (limited by block nets) ■ Higher population densities	■ Two passes ■ Use block nets to prevent fish movement. ■ Apply equal effort for each pass ■ 24 h rest between passes to reduce gear avoidance and allow for fish to return to preferred habitats. ■ Collect site-specific habitat data (e.g., width, depth, gradient, thalweg depth, cover type, water velocity, temperature, clarity)	■ Mark-recapture estimates are fairly robust to potential bias introduced from variable habitat (cover %) and crew experience. ■ Time delay between the marking and recapture sampling ■ 24 hour recovery period ■ Leaving the block nets in place overnight

Sources: Lockwood and Schneider 2000, Hubert and Fabrizio 2007, Johnson et al. 2007

Table 3: Equipment Checklist – Fish Sampling

General Items	
Field forms (waterproof)/Tablet	☐
Field maps printed and on field tablets	☐
Camera	☐
GPS with spare batteries	☐
Leak-free chest waders with wading belt	☐
Personal floatation device	☐
Throw bag	☐
Fish collection permit(s)	☐
Flagging tape	☐
Aquaseal (for repairing waders/wetsuits)	☐
Electrofishing	
Backpack electrofisher(s)	☐
Spare electrofisher batteries	☐
Pole mounted anode (and spare)	☐
Cathode	☐
Dip nets	☐
Block Nets	☐
Collapsible Buckets and fish holding bins or live wells (sized appropriately for the fish likely to be captured)	☐
Thermometer and conductivity metre	☐
Polarized sunglasses	☐
Brimmed hat	☐

Linesman gloves (15" length, 5000 volt rating)	☐
Minnow Trapping	
Minnow Traps with clips	☐
Floats/buoys	☐
Rope	☐
Bait (e.g., salmon roe, dried crab/prawn bait, cat food, canned fish etc.)	☐
Snorkelling	
Mask and snorkel	☐
Wading boots and neoprene socks	☐
Fins (can be useful in large systems)	☐
Thermal protection (wetsuit or dry suit), preferably black, blue, or brown coloured.	☐
Neoprene hood and gloves	☐
Fleece undergarments (for drysuit only)	☐
Knee and elbow pads	☐
Handheld underwater light	☐
Waterproof cuff or slates for data recording	☐
Pencils	☐
Handheld counter (optional)	☐
Block Nets (depending on the protocol used)	☐
Fish silhouette with parr marks and spots (to assess visibility)	☐
Thermometer	☐
Measuring tape	☐
Fish Processing Kit	

Measuring board	☐
Fish viewer	☐
Electronic scale(s) and/or a hanging digital scale with a cradle net (more suitable for larger fish)*	☐
Weigh boats	☐
Fish species identification guides	☐
Clip board and pencils	☐
Small dip net	☐
Battery operated aerators	☐
Fish holding bins	☐

Notes:

This list includes only equipment necessary for performing fish sampling and does not include items required for remote work or wildlife safety.

** Electronic scales should be selected based on the expected size of fish to be encountered. Aim to use the scale with the greatest precision possible. For large fish, a hanging digital scale with a cradle net may be a preferable method for measuring weight.*

1.1 Key Terminology

This section briefly covers important concepts and definitions relevant to this protocol.

Catchability

Catchability is defined as the proportion of the fish that are available to be captured that is caught by a defined unit of fishing effort (Ricker, 1975). The catchability of fish is equal to the efficiency of the sampling approach. Variations in catchability decrease the accuracy of CPUE estimates as indices of abundance. Catchability can vary with size, sex or other characteristics of fish, as well as with time of day, season, sampling site, water temperature, dissolved oxygen, or other environmental features that affect the ability of the gear to capture fish or influence the distribution of fish relative to the gear (i.e., availability; Hubert and Fabrizio 2007).

Catch-per-unit-effort (CPUE)

The number of fish captured by a particular gear with a particular amount of effort is termed catch-per-unit-effort (CPUE). CPUE is defined mathematically as:

$$CPUE = C/f = qN$$

Where C is the number of fish caught, f is the unit of fishing effort expended, q is the catchability coefficient or probability of catching an individual fish in one unit of effort, and N is the absolute abundance of fish available to the particular fishing effort f .

Units of fishing effort (f) may include individual sets or hauls of gear, the volume or area of habitat sampled, or the temporal duration of sampling (Hubert and Fabrizio 2007). For example, effort with seine nets is often quantified relative to the area or length of shoreline over which the net is pulled.

For electrofishing samples, CPUE is often described in terms of the number of fish caught in a given amount of time (minutes or hours) or length of shoreline sampled (Hubert and Fabrizio 2007).

Sampling Efficiency

The ability of the method to catch the fish that are actually there.

2. SAMPLING STANDARDIZATION

Standardized sampling is defined as sampling with identical gear during the same season (or set of environmental conditions) in the same manner over time or among fish populations (Pope et al. 2010). Standardized sampling does not eliminate bias, but theoretically holds the bias constant so that differences in indices (e.g., CPUE) computed from samples among years or fish populations can be attributed to relative changes in a population or relative differences among populations (Pope et al. 2010). Standardization of fisheries data can provide several benefits including:

- Data collected over a period of time can be compared easily (e.g., standardized time series data can be used to assess the success or failure of management actions; Bonar et al. 2009).
- Data can be compared across large geographic regions (e.g., standardized sampling over large regions can provide sufficient sample sizes to test the effects of habitat improvements, regulations, or management actions; Bonar et al. 2009)
- Standardization can encourage data sharing and production of large-scale data sets for future assessments (Bonar et al. 2009).

Standard sampling requires (1) fishing the same gear types in the same manner and (2) using a sampling design that allows data to be compared among surveys (Bonar et al. 2009). These are discussed briefly in the sections below.

2.1 Gear Selection

All fishing gears have inherent biases. For example, minnow traps will target a different group of species and length intervals than electrofishing.

Active sampling methods such as electrofishing, snorkelling, and seining capture a portion of the true fish population or community at a discrete location over a relatively short time interval (Peterson and Paukert 2009). Conversely, passive sampling methods such as minnow traps are typically fished over a longer period of time and rely on fish movement for fishes to be captured. Given that passive methods rely on fish movements, it is difficult to define the precise area that is being sampled. This can complicate comparisons of active and passive gear catches (Peterson and Paukert 2009). Data conversions between gears can lead to increased variance and model error which can make it more difficult to assess trends of a fish population (Peterson and Paukert 2009). For this reason, in standardized sampling programs, it is best to be consistent and avoid changing sampling gears.

Table 4 provides a summary of commonly used fish sampling gear and compares their suitability based on habitat conditions and survey objectives.

Table 4: Summary of fishing gear suitability based on project objectives and habitat conditions (adapted from: Portt et al. 2006)

Fishing Gear	Depth Constraints	Habitat Constraints	Comments
Backpack Electrofisher	Limited to safe wading depth	Low conductivity water, high turbidity or dense vegetation decreases efficiency	Common gear used for many habitats
Seine Netting	Limited to safe wading depth	Limited to habitat with no or sparse structure (smooth bottom); low velocity	Capture efficiency is habitat-dependent
Minnow Trapping	Water depth must be sufficient to submerge the trap entrances	Limited to low velocity habitats	Limited to small fish. Efficacy is species-dependent and often low.
Snorkel Surveys	Water depth must be sufficient to allow the observer(s) to submerge a mask (i.e., at least 20 cm).	Water clarity and structure affect visibility; high current is an impediment	Direct underwater observation can allow for a better understanding of how fish use the habitats available to them.

Sources: Portt et al. 2006, Johnson et al. 2007

2.2 Sampling Design and Site Selection

For unbiased comparisons of fish abundances within habitats over time and/or among different habitats, it is either necessary to exhaustively sample the focal habitats or to *randomly* select sampling sites (Bonar et al. 2009). Random sites should be chosen from a complete set of potential sites that both cover the full area of interest and take into account the particular type of sampling or gear (Smokorowski et al. 2015). If the habitat is relatively uniform, simple random selection is appropriate. In this instance, a predetermined number of sampling sites is selected from all possible sampling sites such that every potential site has an equal chance of being selected (Hansen et al. 2007). However, if there is substantial habitat heterogeneity across locations, then a stratified random sampling approach is appropriate (Smokorowski et al. 2015). Stratified random sampling involves splitting the area into relatively uniform strata based on known or anticipated physical, chemical or biological differences and then randomly selecting sites from within each defined strata (Smokorowski et al. 2015). The strata may be defined by the objective of the restoration project (e.g., focusing on salmon use of deep pools), but it is necessary to also sample from other strata that the fish are able to access to avoid biasing abundance estimates based on fish behaviour or mesohabitat preferences.

Once the monitoring sites are initially selected, there is a choice to either (1) sample the same fixed locations over time, or (2) randomly select sites each time the waterbody is surveyed (Smokorowski et al. 2015; Bonar et al. 2009). Under option (1), it is also necessary to specify whether the 'fixed location' is fixed geographically (e.g., a GPS waypoint) or fixed geomorphically (e.g., at a particular pool, which may move over time due to natural fluvial processes). For understanding the effects of restoration actions, it is generally recommended to use option (1), sampling the same locations over time, to minimise spatial variability and increase the statistical power available to detect the impact of the restoration.

All sites should be well described (using photos and maps) so others can relocate the site(s) and identify the starting and ending points of the survey area(s).

3. FISH ASSESSMENT PROCEDURES

Prior to initiating any fish sampling program, appropriate research and/or sampling permits should be obtained from the appropriate authority. This should be done well in advance of the sampling dates to allow adequate time for processing.

3.1 Backpack Electrofisher

Backpack electrofishing is one of the most widely used methods for sampling fishes. It is well suited to targeting fish species in wadeable watercourses with complex habitats (e.g., undercut banks, woody debris, pools, boulders etc.). Table 5 describes criteria and equipment appropriate for electrofishing in varying stream conditions/sizes. Three common electrofishing methods are described in Table 2 and the step-by-step protocol is presented below.

When sampling with a single backpack-mounted electrofisher unit, a three person crew is ideal to allow for one person to operate the electrofisher, at least one person dedicated to netting fish, and one person to carry a portable live well or bucket and provide backup netting. Under **no circumstances** should electrofishing be performed alone.

Table 5: General criteria and electrofishing equipment appropriate for use in varying stream conditions (Adapted from Temple and Pearsons 2007)

Equipment	Temperature (°C)		Discharge (m ³ /s)		Stream Width (m)		Conductivity (µS/cm)	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
1 backpack Electrofisher	6	18	0	0.3	0	7	40	250
2 backpack Electrofishers	6	18	0.3	3.3	7	15	40	250

Step 1 (for multi-pass removal and mark-recapture only) – Install block nets

- Install block nets at both the upstream and downstream boundaries of the site, perpendicular to the flow of water. Secure the nets to each bank and weight the lead line (i.e., bottom of the net) to the stream bottom with rocks or sandbags
 - Ensure the mesh size of the nets is small enough to prevent movement of the smallest fish that will be sampled.
 - Avoid placing nets where conditions may cause the nets to fail (e.g., swift current).
 - The top of the net should extend at least 30 cm above the water surface to minimize the potential for fish to escape the sample site. Branches or sticks may be used to prop the net up across the channel.
 - The block nets should be checked regularly to both remove debris and monitor for stunned and/or impinged fish.

Step 2 – Place fish holding bins along the stream margin

- Place buckets at 10-25 m intervals along the stream margin to hold captured fish. This allows fish to be returned close to the location from which they were captured and helps to avoid overcrowding.
- Buckets should, where available, be equipped with aerators and the water should be refreshed regularly.

Step 3 – Measure stream temperature and water conductivity

- Measure and record stream temperature and water conductivity.
- Adjust electrofisher settings to the manufacturer's recommended guidelines based on water conductivity and temperature measured in the previous step.

Step 4 – Perform safety checks and begin electrofishing

- Conduct safety check with the electrofishing crew(s) prior to electrofishing.
- Clear or reset the seconds timer on the electrofishing unit and begin electrofishing in an upstream direction.
 - When stream size requires two electrofishing units (Table 5), both crews should move upstream at the same pace, working side by side.
 - All available habitats should be sampled.
- One or two crew members, each equipped with dip nets, should position themselves around the anode to facilitate the capture of stunned fish. One crew member should carry a collapsible bucket during sampling to facilitate the safe and swift transfer of fish to the holding bins on shore.

A large plastic jug with a handle (which can be looped on a wading belt) and a lid can be a useful way to hold/transfer small fish.

- Fish should be transferred to live wells frequently by the netter(s) and should be monitored for signs of stress. Consider adding rocks or leaves to the live wells to provide cover for captured fish. To help prevent predation in the live wells, keep large predatory fish separate from smaller fish.
- Once the upstream end of the sampling boundary is reached, record the total time fished (see seconds timer on the electrofishing unit).

Step 5 – Process fish

- Retrieve live wells and count, identify and measure fish.
- Based on the electrofishing method employed, the next step is as follows:
 - **CPUE sampling:** Release fish close to where they were initially captured.
 - **Multi-pass removal:** Hold the captured fish in live wells until all passes have been completed.
 - **Mark-recapture:** Clip or notch a portion of one of the fins to identify marked fish. Alternatively, mark fish using coloured tags (Appendix A). Release fish close to where they were initially captured and record the release time.

Step 6 (for multi-pass removal and mark-recapture only) – Recovery Period

- **Multi-pass removal:** Avoid disturbing the stream section for at least **1 hour** (to reduce gear avoidance).
- **Mark-recapture:** Allow fish to recover for at least **24 hours**. During this time, avoid disturbing the stream section.

Step 7 (for multi-pass removal and mark-recapture only) – Perform another electrofishing pass

- **Multi-pass removal:** Electrofish the sampling site again following the same process described above in steps 2 – 4. Apply an effort similar to that of the first electrofishing pass.
- **Mark-recapture:** Electrofish the sampling site again following the same process described above in steps 2 – 4. Apply an effort similar to that of the first electrofishing pass.

Step 8 (for multi-pass removal and mark-recapture only) – Process fish

- Count, identify and measure captured fish.
 - **Multi-pass removal:** Hold the captured fish in live wells until all passes have been completed (repeat Steps 6-7 as necessary). Once all passes and fish processing is complete, fish can be released near their point of capture after fully recovering from handling.
 - **Mark-recapture:** Pay special attention to identify and enumerate marked fish and unmarked fish on the field data sheets. Once processing is complete, fish can be released near their point of capture after fully recovering from handling.

Step 9 (for multi-pass removal and mark-recapture only) – Remove block nets

- Remove block nets at the upstream and downstream site boundaries.
-

3.2 Seine netting

Seining is a technique well suited to capturing fish in relatively shallow, slow-moving water with few obstructions (e.g., boulders, large woody debris (LWD), dense aquatic vegetation etc.), in areas where fish concentrations are high. This method can be applied to sample relatively large areas over a short period of time. Seine nets are typically weighted on the bottom (lead line) and have buoys on the top (float line), in this way, the net remains vertical when pulled through the water column. Often, a bag is built into the midpoint of the net, so that fish move into the bag as the net is pulled through the water. Seine nets can be deployed in a variety of ways depending on the target species, water depth, habitat, and currents. A common method for seine netting in wadeable streams is briefly described below:

1. Prior to seine net deployment, mark the float line in 1 m increments so that the distance from shore can easily be noted.
2. Stack the seine net on the beach such that it can unravel neatly with the lead line oriented under the float line.
3. Two crew members start on shore, each holding a pole or stick attached to opposite ends of the seine. One crew member pulls one end of the seine straight out from shore until the net is fully extended or the water is too deep to safely wade. Each pole is held vertically or slightly angled (with the lead line ahead of the float line) to keep the lead line in constant contact with the substrate. The nearshore end of the seine is kept in very shallow water or slightly on shore.
4. Both crew members pull the seine along the shore for some distance and then the end furthest from shore is pulled in an arcing, semi-circular motion back to shore, keeping the net as elongated as possible (Figure 1).
5. Measure and record the area sampled in order to express the number of fish captured per unit area.

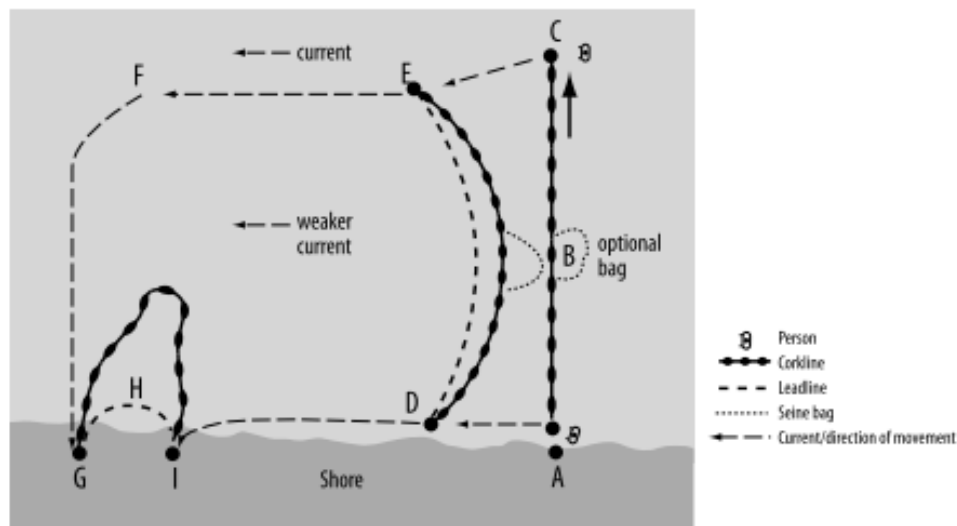


Figure 1: Illustration of beach seine deployment and retrieval. Note, the distance from A-I can be several hundred metres (Source: Hahn et al. 2007).

3.3 Minnow Trapping

Gee minnow traps are typically used to capture juvenile (e.g., salmonid fry) and small adult fish in low velocity streams or littoral habitats. Water depth must be sufficient to submerge the trap entrances. The traps consist of two metal mesh cylinders locked together with a clip and attached to a rope and buoy. Traps are baited prior to deployment at sampling sites, and are often left to fish overnight and retrieved the next day. All traps **must** be retrieved as they continue to fish indefinitely. Typical minnow trap gear specifications are shown in Table 6. When sampling paired restoration and control/reference sites, the same bait type and amount should be used to minimise variability in catch data, as well as recognising where bait may attract fish from nearby habitats that are outside of the study area.

Table 6: Typical Minnow Trap Gear Specifications

Feature	Measurement
Diameter	23 cm
Length	42 cm
Trap mesh size	6.4 mm
Trap opening	2.5 cm
Catch per unit effort (CPUE)	Fish per 24-h

3.4 Snorkel Surveys

Snorkel surveys are widely used to assess abundance, distribution, size structure and habitat use of salmonids. Benefits and limitations of applying this method are presented in Table 7:

Table 7: Snorkel Survey Benefits and Limitations

Benefits	Limitations
■ Limited disturbance to the target species and limited potential for fish injury. ■ Typically low cost and time-efficient. ■ Snorkelling is often feasible at sites where other methods may not work (e.g., deep, clear water with low conductivity). ■ The limited amount of equipment required is ideal for remote field locations where site access is difficult and it may be challenging to apply other more equipment-intensive methods (e.g., electrofishing, trapping, netting). ■ Direct underwater observation can allow for a better understanding of how fish use the habitats available to them.	■ Smaller fish and bottom-dwelling fish may be more difficult to detect/count. ■ Differences in fish behaviour and the amount of cover available can affect the accuracy of counts ■ Surveyors may misidentify fish, double-count fish, or fail to detect fish. ■ Fish length is estimated, not measured directly and fish cannot be weighed or tagged. ■ Variability among observers' in data collection may contribute to inconsistencies or errors.

Snorkel survey techniques vary depending on project objectives and site conditions. For snorkel surveys to be effective, site conditions must meet minimum criteria for water clarity, water depth, and water temperature. These criteria are described below:

- **Water clarity** should be sufficient to allow observers to see the stream bottom in the deepest habitats, identify fish by species, and detect fish avoidance behaviour. Observers should measure and record the visibility of a known object (e.g., fish silhouette with parr marks and spots) prior to sampling. To do this, a surveyor should approach the silhouette underwater until

the parr marks are clearly visible and then move away until the parr marks cannot be distinguished. The average of these two distances provides the water visibility.

- **Water depth** throughout the sampling area must have sufficient depth to allow the observer(s) to submerge a mask (i.e., at least 20 cm).
- **Water temperature** influences fish behaviour and may contribute to biased underwater counts. For example, studies have demonstrated that stream-dwelling trout and salmon typically become more cryptic at temperatures less than 9°C, making them more difficult to detect in snorkel surveys. If sampling must be focused on colder seasons, consider sampling at night, no sooner than 1 hour after sunset.

Once it has been established that site conditions are suitable for performing snorkel surveys, crews can proceed with the protocol outlined below.

Step 1 – Record general site observations

- Record reach location information, names of all survey team members, as well as weather conditions, water temperature, and visibility (refer to guidance described above).
- Surveyors should carefully assess the potential hazards within the sampling reach before entering the water.

*Step 2 – Enter the water at the downstream extent of the site to be surveyed **

- Observer(s) enter the water downstream from the area to be sampled and pause to allow fish to resume normal behaviour after the initial disturbance. Most salmonids orient into the current; for this reason, observers moving upstream are less likely to startle fish.

*In larger systems where upstream counts may not be possible, snorkelers may conduct counts floating downstream, remaining as motionless as possible. Snorkelers can hold onto a length of PVC pipe to maintain equal distance and pace with other snorkellers (Figure 2).

Step 3 – Begin the snorkel survey

- Observer(s) move slowly upstream, avoiding sudden movements and taking care not to disturb the substrate. The survey method will vary depending on the characteristics of the habitat:
 - In small streams, where an observer can see clearly from bank to bank from a single point the observer proceeds in a zigzag pattern between the banks.
 - Where water depth, turbulence, or clarity limit the ability to see, the observer first moves up one bank and counts all fish to the limits of visibility and then repeats the process for the opposite bank. Alternatively, if additional observers are available, the site can be divided into lanes of equal width and observers can count all fish within their assigned lane (Figure 2).
 - General guidelines for surveys with more than one observer:
 - Lane boundaries should be discussed and clearly defined in advance of the survey and could depend on both site characteristics (e.g., visibility, cover) as well as the abilities of each observer.
 - The distance between observers should not exceed the maximum underwater visibility.
 - Observers should start and stop at the same time, move at the same speed, and remain within their assigned lanes.
 - Observers must take care to avoid counting fish that move among lanes to prevent counting the same fish twice.
 - Observer(s) should thoroughly search all areas within the sample reach, including stream margins and all cover (e.g., substrate interstices, woody debris, undercut banks etc.). A handheld
-

underwater light can be useful to detect concealed or less conspicuous fish. When using lights, disturbance can be minimized by not shining light directly on the fish, by directing lights to the underside of water surface, or by using red filters.

- Target fish are counted by the observer(s) and recorded by species and length-class. To estimate length, a scale such as a ruler or marked dive slate can be placed within the field of view of the diver near where fish are to be observed. Alternatively, measuring tape can be fixed to the diver's glove to use as a calibration tool. Snorkelers can practice estimating fish length by viewing wooden dowels of known length underwater. Training and practice **prior to surveys** can greatly improve the observers' accuracy in length estimation.

Step 4 – End the snorkel survey

- Once reaching the upstream extent of the surveyed area, observer(s) can exit the water, check over the data recorded on the slates and make note of any factors potentially influencing the efficiency of the snorkel survey. This will assist with the interpretation of snorkelling results and provide opportunities for calibration of snorkel counts.
- After completing the survey, observers should measure the surface area of the snorkelled unit. The total length of the unit should be recorded as well as the width at three or more equally spaced intervals. The surface area can be estimated either by multiplying the length of the snorkelled unit by average width or by calculating the area of individual segments and pooling them for a total area estimate. Maximum pool depth should also be recorded if additional habitat surveys are not being conducted.

Step 5 (optional) – Recovery period and survey replication

- Precision (a measure of the repeatability of measurements) can be estimated by replicating the survey. This should be done after fish have had time to recover from disturbance but before conditions that would affect the survey results change (e.g., light levels, turbidity).
- Repeat Steps 1-4

A note on assessing the accuracy of snorkel-based estimates

The relative accuracy of snorkel estimates varies from stream to stream, for this reason, snorkel counts should be periodically calibrated with other methods of estimating population size. This can be done by comparing snorkel counts with abundance estimates derived from other methods (e.g., electrofishing). One way to do this involves combining snorkelling with other fishing techniques to derive mark-recapture population estimates.

In practice, this requires the system to be closed off with block nets (refer to protocol described in Section 3.1). The mark-recapture method assumes (1) no immigration or emigration occurs throughout the survey (2) marking does not affect mortality, and (3) both marked and unmarked fish are randomly mixed and have equal chances of being seen. Fish can then be captured, marked (with coloured tags or dyes with colour coding patterns), and released back to the sample area (refer to protocol described in Section 3.1). After a recovery period, snorkel surveys can be conducted in the same reach to count marked and unmarked fish by species and size class. The ratio of the number of fish resighted (i.e., those with marks that are counted) can be used as a calibration factor for the effectiveness of a snorkel survey compared to other survey methods.

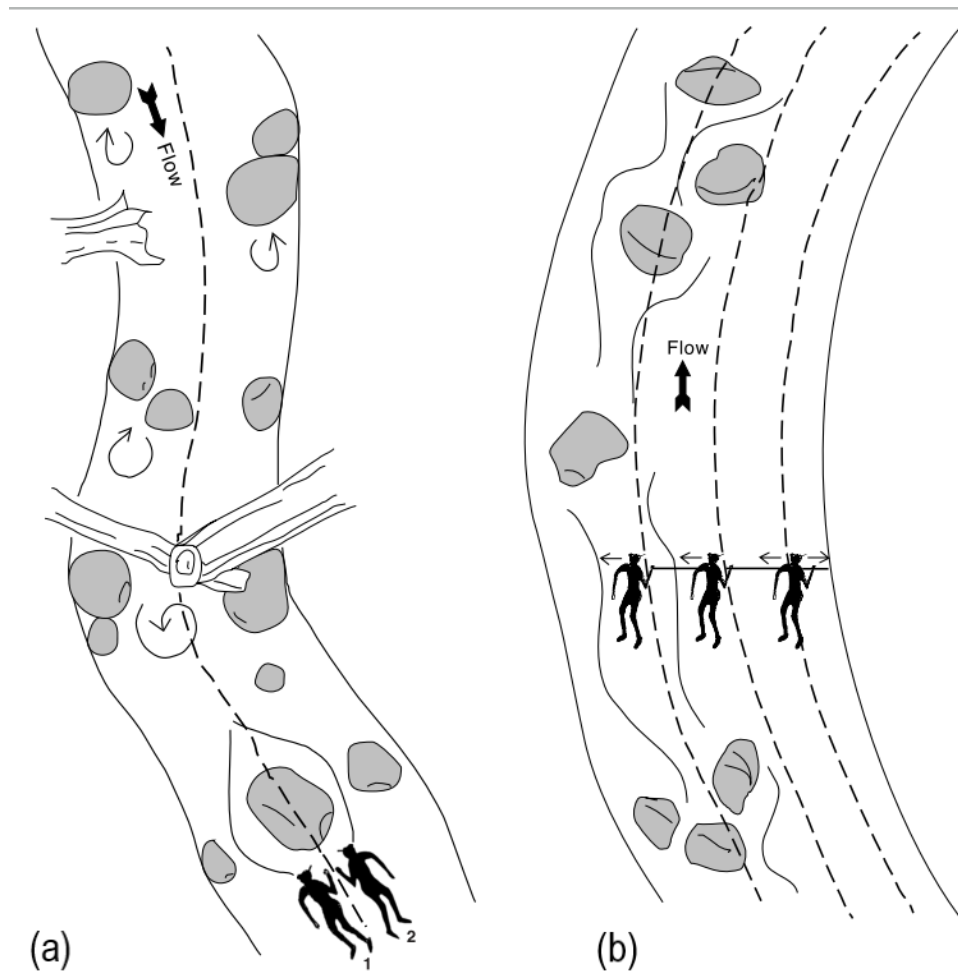


Figure 2: Illustration of snorkel surveys with more than one observer (Source: Thurow 1994).

(a) two snorkelers counting fish while moving upstream. Observer 1 counts all fish to the left of centre and observer 2 counts all fish to the right of centre.

(b) Multiple snorkellers maintaining equal distance and pace with a pole as they move downstream in a large river. The unit is equally divided and fish are counted as the snorkellers pass them. The small horizontal arrows indicate the directions each snorkeler counts fish and the dashed lines represent the approximate paths of the snorkelers.

4. FISH HANDLING PROCEDURES

4.1 Fish holding

After capture, fish should be held in bins or live wells that are sized to allow for the fish to swim freely. The water should be replaced regularly and battery powered aerators can be used within fish holding bins to ensure fish are receiving adequate oxygen while being held. Live wells (essentially floating netted pens), allow water to flow freely through the netted enclosure, greatly reducing oxygenation issues and creating a less stressful environment for fish. These can be purchased or built with materials available in most hardware stores.

4.2 Basic Measurements

Basic measurements should be taken from all fish even if they are not the target species. Refer to Table 3 for a list of fish sampling equipment. Measurements should be taken for all fish according to the steps outlined below:

1. With bare hands remove the fish from the holding bin and place it on the measuring board on the right side of the body. Note, aquarium dip nets can be helpful to remove individual small fish from holding bins.
2. With the tip of the snout flush with the end of the measuring board, measure either fork length or total length and record the value.
 - a. **Fork length** is measured from the tip of the snout to the middle caudal fin ray. This measure is suited to fish that have forked caudal fins, and is recommended for salmon (Figure 3)
 - b. **Total Length** is measured from the tip of the snout to the tip of the longest caudal fin ray when the lobes of the tail are pressed together. This measure is mainly used on fish that have uneven caudal fin lobes or on fish without forked tails (e.g., burbot, sculpin, lamprey, etc.; Figure 3).
3. Transfer the fish to the scale and record the weight.
4. Transfer the fish to a holding bin for recovery.

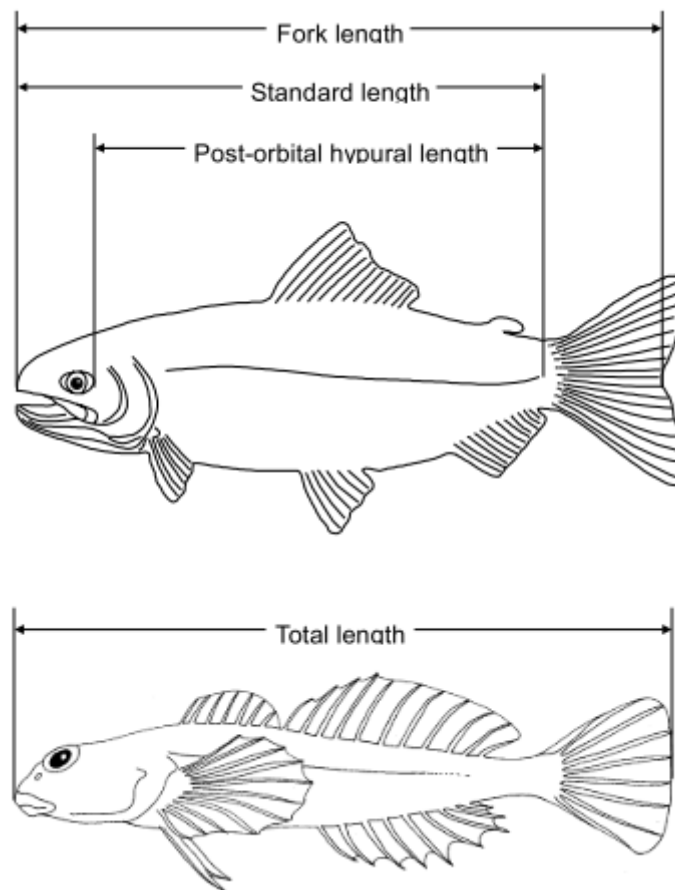


Figure 3: Illustration of commonly used metrics for measuring fish length (Source: RIC 1997)

5. REFERENCES

- Bonar, S. A., S. Contreras-Balderas, and A. C. Iles. 2009. An Introduction to Standardized Sampling. Pages 1–11 in S. A. Bonar, W. A. Hubert, and D. W. Willis, editors. Standard methods for sampling North American freshwater fishes. American Fisheries Society, Bethesda, Maryland.
- CBFWA (Columbia Basin Fish and Wildlife Authority PIT Tag Steering Committee). 1999. PIT tag marking procedures manual. CBFWA. Portland. 26p.
- Fisheries and Oceans Canada (DFO). 2004. Marking and Tagging of Finfish. Canada Department of Fisheries and Oceans Animal-Use Training Template. Available at: https://ccac.ca/Documents/Education/DFO/6_Marking_and_Tagging_of_Finfish.pdf
- Hahn, P.K.J., R.E. Bailey, and A. Ritchie. 2007. Beach Seining. Pages 267–324 in D.H. Johnson, B. M. Shrier, J.S. O'Neil, J.A. Knutzen, X. Augerot, T.A. O'Neil, and T.N. Pearsons, editors. Salmonid field protocols handbook: techniques for assessing status and trends in salmon and trout populations. American Fisheries Society, Bethesda, Maryland.
- Hansen, M. J., T. D. Beard Jr., and D. B. Hayes. 2007. Sampling and experimental design. Pages 51–120 in C. S. Guy and M. L. Brown, editors. Analysis and interpretation of freshwater fisheries data. American Fisheries Society, Bethesda, Maryland
- Hawkins, J.A. 2005. Standard Operating Procedure Floy® Tagging fish. Larval Fish Laboratory, Department of Fishery and Wildlife Biology, Colorado State University.
- Hayes, D. B., J. R. Bence, T. J. Kwak, and B. E. Thompson. 2007. Abundance, biomass, and production. Pages 327–374 in C. S. Guy and M. L. Brown, editors. Analysis and interpretation of freshwater fisheries data. American Fisheries Society, Bethesda, Maryland, USA.
- Hubert, W.A. and Fabrizio, M.C. 2007. Relative abundance and catch per unit effort. Analysis and Interpretation of Freshwater Fisheries Data (eds C.S. Guy and M.L. Brown), pp. 279–325. American Fisheries Society, Bethesda, Maryland, USA.
- Johnson, D. H., B. M. Shrier, J. S. O'Neil, J. A. Knutzen, X. Augerot, T. A. O'Neil, and T. N. Pearsons. 2007. Salmonid field protocols handbook: techniques for assessing status and trends in salmon and trout populations. American Fisheries Society, Bethesda, Maryland.
- Lockwood, R. N., and Schneider, J. C. 2000. Stream fish population estimates by mark-and-recapture and depletion methods. Manual of Fisheries Survey Methods II: With Periodic Updates, January, 1–14.
- Peterson, J. T., and C. P. Paukert. 2009. Converting Nonstandard Fish Sampling Data to Standardized Data. Pages 195–212 in S. A. Bonar, W. A. Hubert, and D. W. Willis, editors. Standard methods for sampling North American freshwater fishes. American Fisheries Society, Bethesda, Maryland.
- Pine, W. E., J. E. Hightower, L. G. Coggins, M. V. Laretta, and K. H. Pollock. 2012. Design and analysis of tagging studies. Pages 521–572 in A. V. Zale, D. L. Parrish, and T. M. Sutton, editors. Fisheries techniques, 3rd edition. American Fisheries Society, Bethesda, Maryland
- Pope, K.L., Lochmann, S.E. and Young, M.K. 2010. Methods for assessing fish populations. Inland Fisheries Management in North America, 3rd edn (eds W.A. Hubert and M.C. Quist), pp. 325–352. American Fisheries Society, Bethesda, MA.
- Portt, C.B., G.A. Coker, D.L. Ming, and R.G. Randall. 2006. A review of fish sampling methods commonly used in Canadian freshwater habitats. Can. Tech. Rep. Fish. Aquat. Sci. 2604 p.
- Resources Inventory Committee (RIC). 1997. Fish Collection Methods and Standards – Version 4.0. Prepared for Resource Inventory Committee. Victoria.
-

- Ricker, W.E. 1975. Computation and interpretation of biological statistics of fish populations. Bulletin 191. Fish. Res. Board Can. 382 p.
- Smokorowski, K.E., Bradford, M.J., Clarke, K.D., Clément, M., Gregory, R.S., Randall, R.G. 2015. Assessing the effectiveness of habitat offset activities in Canada: Monitoring design and metrics. Can. Tech. Rep. Fish. Aquat. Sci. 3132: vi + 48 p.
- Temple, G. M., and T. N. Pearsons. 2007. Electrofishing: backpack and drift boat. Pages 95–132 in D. H. Johnson, B.M. Shrier, J.S. O'Neal, J.A. Knutzen, X. Augerot, T.A. O'Neil, and T. N. Pearsons. Salmonid field protocols handbook—techniques for assessing status and trends in salmon and trout populations. American Fisheries Society, Bethesda, Maryland
- Thurrow, R. F. 1994. Underwater methods for study of salmonids in the Intermountain West. U.S. Forest Service, Intermountain Research Station, General Technical Report INT-GTR-307, Odgen, Utah
- Quist, M.C., K.I. Bonvechio, and M. S. Allen. 2009. Statistical Analysis and Data Management. Pages 171–190 in S. A. Bonar, W. A. Hubert, and D. W. Willis, editors. Standard methods for sampling North American freshwater fishes. American Fisheries Society, Bethesda, Maryland.
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APPENDIX A: TAGGING METHODS

Tagging data can be used to assess fish movement and estimate growth rates, demographic parameters and abundances. The tagging method used will depend on the project objectives.

Basic considerations for selecting a particular type of tagging method include:

- Potential effects on fish survival, behaviour and growth
- Tag permanency and recognition
- Number and size of fish
- Stress of capture, handling, and marking
- Total costs
- Recovery of marked fishes

Two commonly used tagging methods are described below.

T-Bar Anchor Tags (Floy® Tags)

Floy tags, also known as T-Bar anchor tags, are composed of an internal attachment end (T-shaped) embedded into the body of the fish and an external coloured plastic tube imprinted with study-specific information (e.g., unique identification number, agency etc.). These tags are commonly used on a wide variety of fishes as they are relatively easy to attach, and readily visible. Tag retention relies on the proper placement of the tag behind the pterygiophores (i.e., the bony elements by which rays of the fin are supported) or other rigid body structure to provide a firm attachment point (Figure A-1).

The general procedure described below is adapted from Hawkins (2005). Materials include: a tagging gun with sharpened stainless steel needles, Floy T-bar anchor tags, 70% ethanol, and a diamond needle sharpener. Fish must be handled with care during tagging to minimize stress.

1. Cleanse needles and tags in 70% ethanol for 10 minutes.
 2. Hold the fish upright with the head and body underwater (if possible) and the dorsal surface raised out of the water. Take care not to restrict opercula movement.
 3. Note the tag location, which should be on the left side of the fish, below the dorsal fin and between the second and third dorsal fin ray. The tag should trail to the posterior at an acute angle to the body axis (to reduce drag). A couple of scales should be carefully removed with the needle tip prior to puncture to facilitate entry of the needle.
 4. Insert the needle into musculature, pushing it through the dorsal pterygiophores past the midline. The needle should be angled towards the front of the fish but behind the fin ray (do not insert the tag perpendicular to the fish's body). Do **not** penetrate the opposite side of the fish.
 5. Compress the trigger to insert the tag. Rotate the tag gun 90 degrees with the trigger depressed to dislodge and lock the tag behind the pterygiophores.
 6. Release the trigger and remove the needle from the fish. Gently tug on the tag to set and test the anchor.
 7. Record the tag number and colour.
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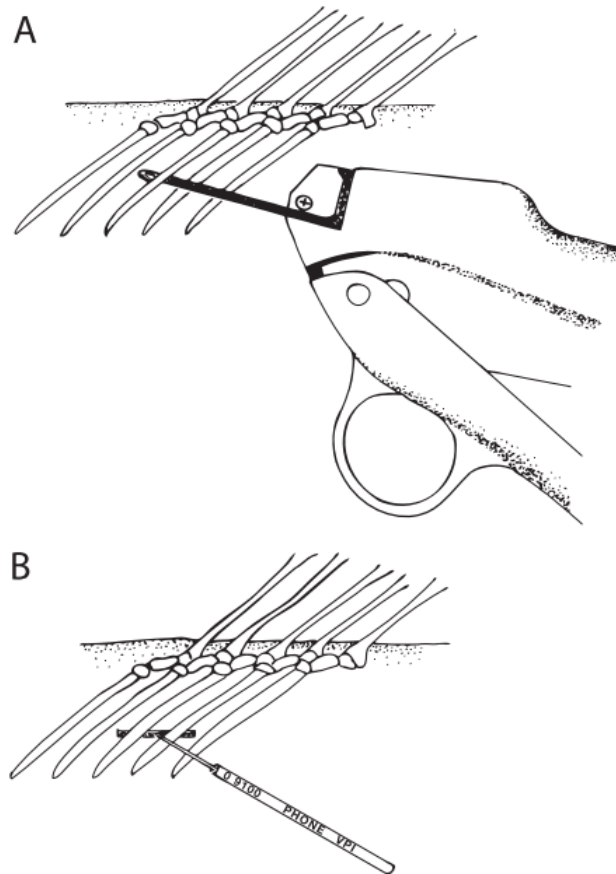


Figure A-1: Illustration of (A) placement of tagging gun needle and (B) tag positioned between the pterygiophores of fish (Source: Pine et al. 2012)

Passive Integrated Transponder (PIT) Tags

PIT tags are small, glass-coated microchips that can be activated and detected at close range by a specialized scanner (also known as a reader). PIT tags can last for the lifetime of a fish as they do not require a power source and are small enough to be injected into juvenile fish. Handheld scanners can be used to check for tags in fish that have been captured; alternatively, readers can be installed into antenna arrays to detect any tagged individuals that swim by. Multiple antennas can be used to determine the direction of fish movement. PIT tags may offer an advantage over physical identification tags as they can be detected remotely, meaning recapture of tagged fish is not required. It is recommended that fish are at least 65 mm in fork length before being PIT tagged (DFO 2004; CBFWA 1999).

The general procedure described below is adapted from Hawkins (2005), DFO (2004), and CBFWA (1999). Materials include: PIT tag reader, PIT tags, 70% ethanol, PIT tag injectors (12 gauge needle modified with stainless steel plunger).

1. Sterilize the tagging needle and injector in 70% ethanol for 10 minutes prior to tagging, as well as in between fish.
2. Load a PIT tag into the barrel of the hypodermic needle.
3. If the fish is small enough to be held comfortably in hand, the fish should be positioned with the belly facing upward, the caudle fin pointing forwards towards the index finger, and the head in contact with the hand. If the fish is too large to be held comfortably in hand, it can be held at an angle with its head underwater and belly and tail out of the water. Take care not to restrict opercula movement.

4. Note the tag insertion location, on the ventral body wall, midway between the posterior tip of the pectoral fin and the anterior point of the pelvic girdle. The needle should be positioned 1-2 mm off the ventral midline (Figure A-2).
5. Position the needle at a 45 degree angle with the bevel open towards the belly of the fish so the point of the needle is away from the internal organs. Insert the needle no more than 2-3 mm (for fish up to ~ 150 mm) and inject the tag.
6. Vetbond Tissue Adhesive can be applied to the site of the PIT tag insertion to ensure the wound does not gape.
7. Use the handheld scanner to read, record and verify the PIT tag code.

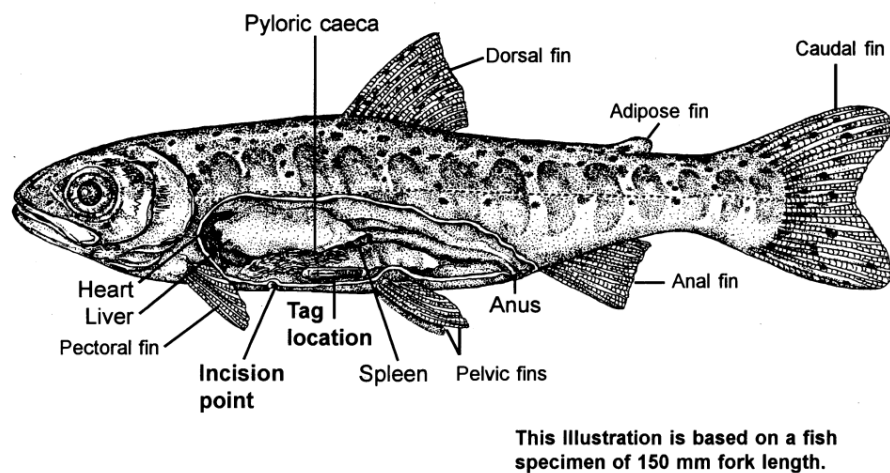


Figure A-2: PIT tag location within the body cavity (Source: CBFWA 1999)

APPENDIX B: FISH SAMPLING FIELD FORMS

[illegible]

Species Codes:
 CH = Chinook, CM = Chum, CO = Coho, PK = Pink, SK = Sockeye, BT = Bull Trout, DV = Dolly Varden, CT = Cutthroat Trout, CCT = Coastal Cutthroat Trout, RB = Rainbow Trout, NSB = Ninespine Stickleback, TSB = Threespine Stickleback, CC = Sculpin (General), CAS = Prickly Sculpin, CCG = Slimy Sculpin

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[illegible]

Stream:				Site:				Upstream Latitude:	
Survey Area (m ²):				Survey Date (d/m/y):				Upstream Longitude:	
Water Temperature:				Survey Crew:				Downstream Latitude:	
Time In:				Visibility:				Downstream Longitude:	
Time Out:				Weather:					
				Maximum Pool Depth:					
Length Class (mm)	Species Observed (Refer to Codes) ¹							Comments	
	CH	CO	RB						
0 to 50									
51 to 80									
80 to 100									
100 to 120									
120 to 150									
150 to 200									
200 to 250									

¹Species Codes:
CH = Chinook, CM = Chum, CO = Coho, PK = Pink, SK = Sockeye, BT = Bull Trout, DV = Dolly Varden, CT = Cutthroat Trout, CCT = Coastal Cutthroat Trout, RB = Rainbow Trout, ST = Steelhead
NSB = Ninespine Stickleback, TSB = Threespine Stickleback, CC = Sculpin (General), CAS = Prickly Sculpin, CCG = Slimy Sculpin